

ECCB: Department of Ecology and Conservation Biology

ECCB 689 Modeling Ecological Niches and Species' Distributions

Term Project Data Plan

Purpose

The data plan should explain what occurrence and environmental data you intend to use, why those data are appropriate for your biological question, and how you will make them compatible for modeling. You may revise this plan as you learn more about your data. Your choices should support an ecological niche or species distribution modeling question. A list of datasets or variables is not sufficient. Explain what each dataset contributes to the analysis. Use the following format.

Required content

1. Research question and model purpose

State your revised research question and identify what you want the model to estimate or compare. Some comparison examples include:

- Environmental conditions associated with occurrences
- Relative environmental suitability within a study region
- Potential changes under another climate period
- Transfer of a model between regions or time periods
- Robustness across alternative assumptions

Briefly explain the biological or conservation value of the question.

2. Occurrence data

Describe your proposed occurrence dataset:

- Species or biological system
- Data source and citation or link
- Geographic and temporal coverage
- Approximate number of records available
- Planned filters or quality-control decisions
- Important sampling biases or limitations

Explain why the records are appropriate for your question and time period.

3. Accessible area and background

Describe your current hypothesis about **M**, the area accessible to the species during the relevant period. Explain:

- How you will define M geographically
- What biological, geographic, or historical reasoning supports that definition
- How background locations will be sampled within M
- One alternative definition of M that you could test

4. Environmental predictors

Identify approximately **three to six candidate predictors**. For each predictor, provide:

- Variable name
- Data source and version
- Units
- Spatial resolution
- Time period represented
- Biological reason for including it
- Whether it is a direct, indirect, or resource variable

Do not select variables because they are readily available. Explain how each predictor could influence the organism or distribution.

5. Data compatibility and processing

Explain how you will evaluate and address:

- Coordinate reference systems
- Spatial extent and raster alignment
- Spatial resolution
- Temporal compatibility between occurrences and predictors
- Missing data
- Correlation among predictors

Identify any resampling, cropping, masking, or transformation you expect to perform.

6. Feasibility and uncertainty

Identify one data problem or modeling limitation you expect to run into and explain how you might address it. Also, include any questions here on which you would like feedback.

Format and submission

- Due Friday, October 2 by 11:59 PM
- Up to three pages, single space
- PDF format
- 12-point Times New Roman
- Tables may be used for environmental predictors
- Include citations or links for all proposed datasets
- Submit through the Data Plan assignment in Canvas